13. LIMITATIONS OF LOW-CODE PLATFORMS FOR A COMPLEX ENTERPRISE USE CASE

Aim

To analyze the technical, architectural, and operational limitations of low-code development platforms when applied to complex enterprise requirements, using the Insurance Claim Management System as the example application.

Theory

Low-code platforms support rapid application development (RAD) for simple and medium-sized business applications. However, complex enterprise systems can encounter limitations in customization, scalability, integration, and long-term platform flexibility.

1. Customization Constraints

Low-code platforms depend on pre-built components, visual builders, and standardized workflows. If the Insurance Claim Management System requires highly specialized claim-assessment algorithms, complex fraud-detection logic, or a fully customized user interface, the visual builder may become restrictive compared with traditional programming.

2. Scalability and Performance Bottlenecks

An insurance organization may receive large numbers of claims, supporting documents, photographs, and videos. High volumes of simultaneous submissions, searches, and payout calculations can create performance bottlenecks or encounter platform resource limits. Performance must be tested against the expected workload.

3. Vendor Lock-in and Proprietary Codebases

Applications built within a proprietary low-code platform may rely on platform-specific forms, workflows, data structures, and scripting. Moving the Insurance Claim Management System to another platform or independent infrastructure may require redesigning and migrating parts of the application.

4. Complex Integration Challenges

An insurance claim application may need to connect with external policy databases, payment gateways, customer relationship management systems, document verification services, or legacy insurance systems. Built-in connectors may not cover every requirement, so custom APIs or additional integration services may be necessary.

Case Study Scenario: Insurance Claim Management at Enterprise Scale

Enterprise Requirement: An insurance company plans to manage claim submissions across multiple branches, process large volumes of supporting evidence, calculate estimated payouts, and integrate with policy, payment, and customer systems.

Identified Limitations During Implementation

1. Customization Wall: Advanced fraud detection and multi-factor claim-assessment rules may exceed the capabilities of standard visual workflows and require custom scripts or external services.

2. Database and Performance Constraints: Large claim records and media uploads can increase storage, processing, and query demands. Platform limits may affect response times when many users submit or review claims simultaneously.

3. Licensing and Scaling Costs: As the number of users, records, integrations, and API requests grows, subscription tiers and usage limits may increase operating costs. The actual cost should be compared with the requirements and available alternatives.

Result

The case study shows that low-code platforms can help build insurance claim workflows quickly, but complex enterprise use may face customization constraints, performance limits, integration challenges, vendor lock-in, and

scaling costs. The appropriate platform should be selected after evaluating workload, integration needs, security, maintainability, and total cost.